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ACRYLAMIDE 50%

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

a. Product name	Acrylamide 50%
b. Recommend use of the chemical	Flocculant, Paper Strengthen, Adhesive
c. Restriction on use	No data available
d. Manufacture Information	
Factory	Yongsan Mitsui Chemical, Inc. 32, Sapyeong-ro(Bugog-dong), Nam-gu, Ulsan, Korea 052 - 226 - 8861 / 5 Safety Environment Team
HEAD OFFICE	02-3274-9181

2. HAZARD IDENTIFICATION

a. Hazard-Risk Classification	Acute toxicity(oral):	Category 3
	Skin corrosion/Irritation:	Category 2
	Serious eye damage/Irritation:	Category 2
	Skin sensitization	Category 1
	Germ cell mutagens:	Category 1B
	Carcinogens:	Category 1B
	Reproductive txicants:	Category 2
	Specific target organ systemic toxicity(single exposure):	Category 1 (nervous system, tetis)
	Specific target organ systemic toxicity(repeated exposure):	Category 1 (nervous system, tetis)

b. Label elements including precautionary statements

Symbol



Signal word

Dangerous

Hazard statement

H301	Toxic if swallowed
H312	Harmful contact with skin
H315	Cause skin irritation
H317	May cause an allergic skin reaction.
H319	Cause serious eye damage.
H340	May cause genetic defects.
H350	May cause cancer.
H361	Suspected of damageing fertility or the unborn child.
H372	Causes damage to organ through prolonged or repeated exposure.

Precaution statement

Precaution

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P201	Obtain special instruction before use.
P202	Do not handle until all safety precautions have been read and understood.
P260	Do not breathe dust/fume/gas/mist/vapors/spray.
P261	Avoid breathing dust/fume/gas/mist/vapors/spray.
P264	Wash hands thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.
P272	Contaminated work clothing should not be allowed out of the workplace.
P280	Wear protective gloves/protective clothing/eye protection/face protection
P281	Use personal protective equipment as required.

Response

P302+P352	If on skin, Wash with plenty of soap and water.
P301+P310	If swallowed: Immediately call a poison center or doctor/physician.
P330	Rinse mouth.
P332+P313	If skin irritation or rash occurs, Get medical advice/attention.
P337+P313	If eye irritation persists: Get medical advice/attention.
P321	Specific treatment(see ...on this label)
P312	Immediately call a poison center or doctor/physician.
P308+P313	If exposed or concerned ; Get medical advice/attention.
P362+P364	Take off contaminated clothing and wash if before reuse.
P333+P313	If skin irritation persists ; Get medical advice/attention.
P305+P351+P338	If in eyes : Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continuous rinsing
P314	Get medical advice/attention if you feel unwell

Storage

P405	Store locked up.
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Disposal

P501	Dispose of contents/container to waste management act.
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c. Othe Hazard-Risk which are not included in the classification criteria (NFPA)

Heath	2
Fire	0
Reactivity	2

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical Name	Other name	CAS Number	EN Number	Content (%)
Acrylamide	Vinyl amide	79-06-1	201-173-7	50%
Water	-	7732-18-5	232-791-2	50%

4. FIRST AID MEASURE

a. Inhalation	Move from exposure immediately if adverse effect. If breathing has stopped, apply artificial respiration. Immediately get medical attention.
b. Skin contact	Immediately wash with soap and water for at least 15 minutes.

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- Get medical attention in case of skin disease.
Completely Wash and dry contaminated clothing and shoes before reuse
Thorow away contaminated shoes.
- c. Ingestion
If person is unconscious, turn head to side.
Never make an unconscious person vomit or drink fluids.
Rinse mouth.
Allow vomiting to occur. When vomiting occurs, keep head lower than hips to help prevent aspiration.
Get immediate medical advice/attention.
- d. Eye contact
Irrigate exposed eye with coupious amounts of room temperature water 15 minutes.
Get immediate medical advice/attention.
- e. Note to physician
If ingest chemical, consider gastric lavage or inject activated carbon.

5. FIRE-FIGHTING METHOD

- a. Extinguish media Regular dry chemical, Carbon dioxide, Water spray, Alcohol-resistance foam
- b. Specific hazards arising from the chemical
Pyrosis product Oxides of nitrogen, Oxides of carbon, Ammonia, Corrosive fume.
Fire and explosion hazards Slightly fire hazard.
- c. Specific protective equipment
and precaution for fire-fighter Wear appropriate chemical resistance equipment.
Move container from fire area if it can be done without risk
Cool containers with water spray until after the fire is out.
Stay away from the ends of tank.
Water or foam may cause frothing.

6. Accidental release measure

- a. Personal precaution, protective equipment and emergency procedures
Do not touch spilled material.
Stop leak if possible without personal risk.
Wear appropriate personel protective equipment.
Use adequate ventilation.
- b. Enviornment precaution and protective procedure
Dike for not flowing down the drain, underground water and surface water spills.
Absorb with activated carbon. Collect spilled using mechanical equipment.
- c. Methods and material for containment and cleaning up
Small spills Absorb with sand or other non-combustible material.
Collect spilled material in appropriate container for disposal
Move containers away from spill to a safe area.
- large spills Dike for later disposal
Keep unnecessary people away, isolate hazard area

7. HANDLING AND STORAGE

- Storage Store in cool/dry place
Store in a closed container. Keep away from ingnition source.

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Store in a well-ventillated place.

Handling Do not raise mist, Adequately ventillate to raise mist.

Avoid contact direct with body. Wear appropriate personel protective equipment.

8. Exposure controls and personal protection.

- a. Control parameter (100 % Acrylamide)
- National regulation - TWA : 0.03 mg/m³
- ACGIH regulation - TWA : 0.03 mg/m³
- NIOSH regulation - TWA : 0.3 mg/m³
- b. Appropriate engineerign controls
- Provide lalac exhaust ventillation system. Maintainging an appropriate wind speed.
- Ensure compliance with applicable exposure limits
- c. Personal protective equipment
- Eye protection Wear splash resisitant safety goggles.
- Provide an emergency eye wash fountain and quick drench shower in the immediate work area.
- Body protection Wear appropriate chemical resistance clothing.
- Hand protection Wear appropriate chemical resistance gloves.
- Respiratory protection Under conditions of frequent use or heavy exposure, respiratory protection may be needed
- Respiratory protection is ranked in order from minimum to maximum. Consider warning properties before use.
- Self-contained breathing apparaus.

9. PHYSICAL AND CHEMICAL PROPERTIES

- a. Appearance Liquid
- b. Color Clear
- c. Odor No
- d. Odor threshold Not applicable
- e. pH 6.5 ~ 7.0
- f. Melting point/freezing point Not applicable
- g. Initial boiling point 105.5 °C
- h. Flash point Not applicable
- i. Evaporation rate Not applicable
- j. Flammability(solid, gas) Not applicable
- k. Vapor pressure Similar to water
- l. Vapor density(air=1) Similar to water
- m. Relative density(water=1) 1.036 @ 20 °C
- n. Solubility Soluble @ 25 °C
- o. Auto ignition temperature No data available
- p. Upper/lowew flammability or explosive limits
- p. Viscosity No data available
- q. Partition coefficient No data available

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10. STABILITY AND REACTIVITY

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|----------------------------|--|
| a. Chemical stability | Stable at normal temperature and pressure. |
| b. Condition to avoid | Heating, flames, sparks, exposure to UV |
| c. Incompatible material | Al, acid, alkali, Cu, Strong oxidant, Fe, brass, reductant |
| c. Incompatible material | Thermal decomposition : Ammonia, oxides of carbon, oxides of nitrogen, corrosive |
| d. Hazardous decomposition | May polymerize. May polymerize violently heating above 60 °C or exposure to UV |
| | Closed containers rupture violently |

11. TOXICOLOGICAL INFORMATION

- | | |
|--|---|
| a. Information on the likely routes of exposure; | |
| Inhalation | Harmful if inhaled. Irritating airway. |
| Ingestion | Harmful if swallowed |
| Skin contact | Harmful if absorbed through skin. Irritating skin. |
| Eye Contact | Damage in contact with eye. |
| b. Health Hazard | |
| Acute oral toxic | 248 mg/Kg-Rat LD50 |
| Acute dermal toxic | 3,360 mg/kg-Rabbit LD50 |
| Acute inhalation toxic | 3,000 mg/kg-Rat LD50 |
| Skin corrosive/irritant | Rabbit - Slightly irritation |
| Serious eye damage/eye irritation | Rabbit - Serious damage |
| Skin sensitization | Positive(Guinea pig) |
| Specific target organ toxicity (single exposure) | Low blood pressure, Epileptic seizure, Peripheral neuropathy |
| Specific target organ toxicity (repeat exposure) | Reported dysbasia, defects of memory, speech disorder |
| Genetic toxicity | Positive |
| Toxicity to reproduction | Female animal : Feus reduction, Male : Impregnation reduction |
| Carcinogenicity | |
| IARC | 2A - Group 2A: Possible get cancer (acrylamide) |
| ACGIH, OSHA | A4 - Not classification(insufficient data, Acryl amide) |
| NTP | Possible get cancer (Acrylamide) |

12. ECOLOGICAL INFORMATION

- | | |
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| a. Aquatic and terrestrial ecotoxicity | |
| Fish | mortality NOEC - Lepomis macrochirus - 70 mg/L 96 hr
LC50 - Pimephales promelas (fathead minnow) - 180 mg/L 96 hr |
| Crustacea | mortality NOEC - Daphnia magna (Water flea) - 120 mg/L 48 hr
EC50 - Daphnia magna (Water flea) - 320 mg/L 48 hr |
| b. Persistence and degradability | |
| Persistence | No data available |
| degradability | No data available |
| c. Bioaccumulative potential | |

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Bioaccumulative	Oncorhynchus - 72 h, BCF: 1.65
Biodegradation	70%
d. Mobility in soil	No data available
e. Other adverse effect	No data available

13. DISPOSAL CONSIDERATION

a. Disposal method	Burn waste in incinerators with afterburner. Disposal of wastes in an approved waste disposal facility
b. Disposal precaution	Consideration to precaution all applicable regulation.

14. TRANSPORT INFORMATION

a. UN Number	3426
b. UN proper shipping name	Acrylamide Solution
c. Transport hazard class	6.1
d. Packing group(if applicable)	III
e. Marine pollution	No
e. Special precaution which a user to be aware of or needs to comply with in connection with transport	Avoid direct sunlight. Cold retention measures of container. Reportable Quantity (RQ): 5,000 lbs - U.S
Measures of Leakage	S-B

15. Regulatory information

a. Korea regulations	
Industrial Safety and Health Act	Working environment measurement, Managed hazardous material, Exposure limit set material, Special health examination. Special control material
Chemical control Act	Applicable, Toxic chemical
Dangerous Material Safety Control Act	Not applicable
Wastes Management Act	Not applicable
b. U.S regulation	
TSCA 12(b)	: 5,000 LBS RQ
CERCLA 103 (40 CFR 302.4)	: 1,000 LBS
SARA 302 (40 CFR 355.30)	: 5,000 LBS
SARA 304 (40 CFR 355.40)	: Acrylamide
SARA 313 (40 CFR 372.65)	
SARA hazard category, SARA 311/312 (40 CFR 370.21)	
Acute : Yes	
Chronic : Yes	
Fire : No	
Reactive : Yes	
Sudden release : No	
OSHA regulation (29 CFR 1910.119)	: Not applicable
TSCA	Listed
State Regulation	California proposition 65 : Warning

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EU Regulation

R45 May cause cancer.

R46 May cause heritable genetic damage.

R24/R25 Toxic in contact with skin and if swallowed

R45/23/24/25 Toxic : danger of serious damage to health by prolonged exposure through inhalation,
in contact with skin and if swallowed.

S53 : Avoid exposure - obtain special instruction before use

S45 : In case of accident or if you feel unwell, seek medical advice immediately

16. Other information

a. Information source and reference

Korea Occupational Safety Health Agency MSDS information service [Http://www.kosha.net](http://www.kosha.net)

Croner's: Dangerous Substances.

Sax's Dangerous Properties of Industrial Materials, 12th Ed.

National Institute of Technology and Evaluation, Japan <http://www.safe.nite.go.jp>

HSNO CCID, New Zealand <http://www.ermanz.govt.nz/hs/compliance/chemicals.html>

EU Directive 1999/45/EC

EU Directive 67/548/EEC

European Chemical Substances Information System <http://ecb.jrc.ec.europa.eu/esis/>

EUN Recommendations on the Transport of Dangerous Goods—Model Regulations 16th Ed.

TOXNET, U.S. National Library of Medicine <http://toxnet.nlm.nih.gov>

ECOTOX Database, EPA <http://cfpub.epa.gov/ecotox>

IMDG Code 2008 edition (Amendment 34-08), IMO

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d. Others

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